

PhD Research Proposal

NLP for Colonial Dutch Archives:
Reconstructing Pre-Colonial Indonesian Settlements
from VOC Administrative Records

Prepared for Prof. dr. Suzan Verberne, LIACS, Leiden University

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April 2026

1 Summary

This proposal outlines a PhD project that applies NLP to colonial Dutch administrative archives (1602–1800) to extract structured settlement data for pre-colonial Indonesia, and validates the extractions against an independent geophysical model of volcanic site burial. The project produces three peer-reviewed publications targeting ACL/EMNLP/LREC venues, plus introductory and concluding chapters. It extends LIACS’s “Digging in Documents” and EXALT research lines from modern excavation reports to primary colonial sources—the earliest systematic documentation of Indonesian settlement geography. The PhD is designed as a cumulative thesis (PhD by publication) over 2–3 years of active research.

2 Background and Motivation

2.1 The Archaeological Gap

Conventional archaeological survey in volcanic Indonesia recovers almost none of the expected pre-colonial settlement record. A burial model validated against 363 observed site depths across 7 volcanic systems in East Java (E075) yields Pearson $r = 0.951$ between predicted and observed burial depths. A separate cross-referenced dataset of 51 eruption-site pairs across 12 volcanic systems (E083) confirms median burial depths of 2.5 m at typical settlement locations, with a calibrated mean sedimentation rate of 4.4 mm/yr. Over a 1,000-year pre-colonial window, this places typical sites at depths well below the ~ 2 m threshold of standard archaeological survey. A five-factor cascade model (E133) estimates that $\geq 99.94\%$ of the expected pre-1600 settlement signal in volcanic Java is archaeologically invisible (Amien 2026a).

This is not a theoretical claim. The Philippines, at comparable population density but with non-volcanic (karst) geology, has 4,000+ pre-colonial sites. Volcanic Java has zero confirmed open-air pre-400 CE sites. Population estimation via four independent Monte Carlo methods (E196) yields 1–2 million people at 400 CE, implying a minimum of 694 expected sites at Philippine rates. The taphonomic suppression factor is $\geq 694\times$ (Amien 2026b).

2.2 Why Colonial Dutch Archives?

If the physical archaeological record is inaccessible, reconstruction must come from textual, epigraphic, or remote-sensing sources. The textual source with the highest volume and density is the Dutch East India Company (VOC) archive and its successors (1602–1942). A pilot NLP extraction from 16 volumes of the colonial archaeological service reports (OV, 1912–1929) alone yielded 22,162 structured mentions—including 6,932 named sites, 4,933 administrative locations, and 9,238 material descriptions (E091). A separate Delpher newspaper extraction (E141) recovered 1,768 archaeological records from 46 search queries. The full colonial corpus—spanning dagregisters, scientific journals, technical reports, and newspapers across 340 years—remains largely unmined.

Three characteristics make this archive uniquely valuable:

1. **Scale.** The VOC operated 100+ trading posts across the archipelago for 200 years. Administrative reports, dagregisters, and correspondence contain systematic settlement mentions with geographic specificity.
2. **Incidental stratigraphy.** Colonial infrastructure projects (railways, canals, irrigation, wells) produced thousands of accidental archaeological observations recorded in newspapers, technical reports, and scientific journals—observations free from archaeological selection bias.
3. **Digital accessibility.** Delpher.nl (Koninklijke Bibliotheek) and GLOBALISE (Huygens Institute / VU Amsterdam) have digitised and OCR'd major portions of this corpus, making it computationally accessible for the first time.

2.3 Alignment with LIACS Research

This project is a direct extension of LIACS's archaeological text mining programme:

- **“Digging in Documents” (2017–2021)** developed ArcheoBERTje, a BERT model pre-trained on 658 million words of Dutch archaeological excavation reports, and built an information extraction pipeline for grey literature. The proposed PhD extends this from *modern* excavation reports to *historical* primary sources—the colonial documents that precede modern archaeology.
- **EXALT (2021–2024)** built multilingual semantic search for archaeological archives across English, Dutch, and German. The proposed PhD addresses the same multilingual NER challenge but in a historical register: 17th–18th century Dutch with code-switching to Malay and variable orthography.
- **GLOBALISE** (Huygens Institute, with Piek Vossen's group at VU CLTL) is digitising 5+ million handwritten VOC manuscript pages. The PhD's NER and IE methods are directly applicable to GLOBALISE output, creating a natural collaboration point.

The PhD adds two elements not present in the existing programme: (a) a physically grounded, non-textual validation signal (the VOLCARCH sedimentation model), and (b) a focus on settlement geography rather than artefact description.

3 Research Questions

RQ1: Named Entity Recognition for Historical Dutch

How accurately can a fine-tuned Transformer model identify settlement names, depth measurements, material descriptions, temporal markers, and find events in 17th–19th century Dutch administrative texts, given orthographic variation, OCR noise, and Dutch-Malay code-switching?

RQ2: Temporal Information Extraction and Normalisation

Can temporal expressions in colonial Dutch texts (dates, reign references, relative chronological markers) be extracted and normalised to absolute timelines with sufficient precision to serve as archaeological dating evidence?

RQ3: Historical Place-Name Disambiguation

To what extent can colonial Dutch place names (e.g., *Soerabaja*, *Buitenzorg*) be automatically disambiguated against modern GIS coordinates, and what is the spatial uncertainty of the resulting geocoded records?

RQ4: Physical Validation of Textual Extractions

Do NLP-extracted settlement locations and burial depths correlate with the independent predictions of the VOLCARCH sedimentation model? Specifically: are extracted settlements near volcanoes reported at greater depths, consistent with the calibrated burial rate of 4.4 mm/yr?

RQ1–RQ3 are NLP research questions with contributions to historical NLP methodology. RQ4 is an integration question that connects NLP output to physical science—an unusual validation paradigm for historical text mining, where ground truth is typically textual.

4 Methodology

4.1 Corpus

The primary corpus consists of three source types:

1. **Delpher.nl newspapers and journals** (Koninklijke Bibliotheek): colonial newspapers (*De Locomotief*, *Bataviaasch Nieuwsblad*), scientific journals (*Tijdschrift voor Indische Taal-, Land- en Volkenkunde*), and archaeological reports (*Rapporten Oudheidkundigen Dienst*, 1910–1942). Full-text search and bulk download via KB SRU API.
2. **GLOBALISE VOC transcriptions**: structured text from VOC manuscripts being produced by the GLOBALISE project (5+ million pages in pipeline).
3. **Nationaal Archief**: VOC dagregisters (1624–1799) and administrative correspondence.

4.2 NER for Historical Dutch (RQ1)

Base model. ArcheoBERTje (pre-trained on modern Dutch archaeological texts) or BERTje (Dutch BERT) as starting point. Fine-tune on manually annotated colonial Dutch data.

Entity types. LOCATION, DEPTH, MATERIAL, TEMPORAL, FIND_EVENT, SOIL_CONTEXT.

Challenges. (a) Orthographic variation in colonial Dutch (*ij* → *y*, *oe* → *oo*, variable capitalisation). (b) OCR noise from digitised historical print. (c) Code-switching: Dutch text frequently includes Malay place names, Javanese terminology, and Sanskrit-derived archaeological vocabulary. (d) Domain shift from modern to historical register.

Approach. Spelling normalisation as preprocessing (rule-based + neural), followed by fine-tuned Transformer NER. Data augmentation via back-translation and synthetic OCR noise injection. Evaluation on held-out manually annotated test set (inter-annotator agreement reported).

4.3 Temporal IE and Normalisation (RQ2)

Colonial texts use heterogeneous temporal expressions: absolute dates (“den 14den Maart 1687”), reign references (“ten tijde van Mataram”), relative markers (“oud”, “Hindoesch tijdperk”). The

challenge is normalising these to absolute timelines.

Approach. Rule-based extraction for explicit dates; Transformer-based classification for implicit temporal markers; alignment with the VOLCARCH chronological framework (5 archaeological periods: pre-400 CE, 400–900, 900–1500, 1500–1600, 1600–1942).

4.4 Place-Name Disambiguation (RQ3)

Colonial Dutch used systematic but non-standard romanisation of Indonesian toponyms. Many names have changed or disappeared.

Approach. Three-tier disambiguation: (a) exact match against historical gazetteer (World Historical Gazetteer, HISGIS); (b) phonological fuzzy matching for spelling variants; (c) contextual disambiguation using surrounding geographic mentions and administrative hierarchy (residency → regency → district). Assign spatial uncertainty radius (city: ± 5 km, region: ± 25 km).

4.5 Physical Validation (RQ4)

The VOLCARCH sedimentation model predicts burial depth as a function of distance from volcanic centre, eruption frequency, and elapsed time. NLP-extracted depth measurements from colonial reports provide an *independent* test: if the model is correct, colonial finds near volcanoes should be reported at greater depths than finds far from volcanoes.

Preliminary result. A pilot study (E197, 33 colonial depth records from Delpher, 1870–1941) shows consistency with the VOLCARCH model: Wilcoxon test $p = 0.131$ (non-rejection of model predictions), and a $5.8\times$ enrichment of colonial finds near VOLCARCH-predicted target locations ($p < 0.00001$).

This is, to my knowledge, an unusual validation paradigm for historical NLP: a concrete, physically grounded, non-textual ground truth.

5 Thesis Structure (Cumulative)

6 Timeline

Assuming start date October 2027 or February 2028. Active research period: 2–3 years.

7 Preliminary Results

The feasibility of this project is supported by existing pilot work:

1. **E141: Delpher Colonial Extraction Pipeline** (2026). Rule-based pipeline extracted 1,768 records from Delpher newspaper corpus via KB SRU API across 46 queries. Of these, 165 were geocoded to specific locations and 9 contained archaeological depth measurements. $5.8\times$ enrichment of finds near VOLCARCH-predicted target zones ($p < 0.00001$). This demonstrates that the data exists in newspapers and is computationally extractable.
2. **E197: Colonial Depth Validation** (2026). 33 colonial depth measurements (24 from E091/OV reports + 9 from E141/newspapers) compared to VOLCARCH sedimentation model predictions. Wilcoxon $p = 0.131$: model not rejected. Cross-century validation (colonial observations 1870–1941 vs. pre-colonial sedimentation model).
3. **E091: OV (Oudheidkundig Verslag) NLP** (2026). Rule-based Dutch NLP on 16 OV volumes (1912–1929, 259K lines). Extracted 22,162 structured mentions: 6,932 site

Chapter	Type	Target venue	Content
1	Introduction	—	Problem statement, VOLCARCH motivation, literature review, thesis outline
2	Paper 1 (RQ1)	ACL / EMNLP / LREC	NER for historical colonial Dutch: model, evaluation, error analysis
3	Paper 2 (RQ2–3)	TACL / DSH	Temporal IE + place-name disambiguation: methods, colonial gazetteer, spatial uncertainty quantification
4	Paper 3 (RQ4)	ACL / EACL	Integration: NLP-extracted settlements vs. VOLCARCH predictions. Physical validation of textual extractions
5	Discussion	—	Synthesis, limitations, implications for Indonesian archaeology, future work

Table 1: Proposed thesis chapters. Existing VOLCARCH papers (under review) serve as background cited in Ch. 1, not as thesis chapters.

names, 4,933 administrative locations, 9,238 material descriptions, 742 volcanic references, 26 numeric depth values. Cross-validated against 52 manual entries: 94.2% recall.

4. **Six papers under review** at international venues (EGQJSJ, JCAA, Antiquity, Oceanic Linguistics, Archipel, Archeologia e Calcolatori), establishing the VOLCARCH framework that motivates the NLP application.
5. **arXiv preprint**: arXiv:2604.00023 (phonological substrate detection via ML—demonstrates NLP competence in a heritage linguistics context).

8 Infrastructure and Collaboration

Compute

4× Intel Core i9, 4× NVIDIA RTX 4080, 128 GB RAM (available at UBHINUS, Malang). Sufficient for Transformer fine-tuning and corpus-scale inference.

GLOBALISE

Natural collaboration point. GLOBALISE produces structured VOC transcriptions; our pipeline adds settlement-focused NER and geographic grounding. Piek Vossen’s group (VU CLTL) has expertise in historical Dutch NLP and could provide linguistic validation.

Delpher.nl

Public API, no registration required for public-domain collections. Colonial material pre-1924 is copyright-free.

Leiden libraries

KITLV (Royal Netherlands Institute of Southeast Asian and Caribbean Studies), University Library Special Collections: access to non-digitised colonial materials.

Domain validation

Historian/linguist collaboration for annotation quality and 17th-century Dutch interpretation.

Period	Activities
Months 1–6	Corpus assembly (Delpher API, GLOBALISE liaison). Annotation guidelines + gold standard (500–1,000 sentences). ArcheoBERTje fine-tuning for colonial Dutch NER. Paper 1 draft.
Months 7–12	Temporal IE pipeline. Historical gazetteer construction. Place-name disambiguation evaluation. Paper 1 submission. Paper 2 draft.
Months 13–18	Full pipeline run on complete corpus. Geocoding + spatial overlay with VOLCARCH model. Physical validation analysis. Paper 2 submission. Paper 3 draft.
Months 19–24	Paper 3 submission. Introduction + discussion chapters. Thesis assembly. Defence preparation.
Months 25–30	Buffer for revisions, resubmissions, and defence.

I do not read 17th-century Dutch—this constraint is designed into the methodology: NLP as primary instrument, with human expert validation of outputs.

9 Funding Plan

Primary: BPI Dosen (Beasiswa Pendidikan Indonesia). Indonesian government scholarship for permanent university lecturers. Age limit ~57 (candidate is 48). Funds up to 4 years abroad; 2–3 years claimed for this project. Application cycle: quarterly (March, June, September, December). A Letter of Acceptance from Leiden would materially strengthen the application.

Secondary options:

- NWO / ERC / Horizon Europe-funded positions in the LIACS group, if available within the relevant timeframe.
- MSCA Doctoral Networks (deadline November 2026; no age limit). If Leiden participates in a consortium relevant to heritage NLP, this could serve as an alternative funding vehicle.

Logistics. Family and university clearance obtained for full relocation to Leiden for the research period. Flexible on residency pattern (full relocation or periodic research visits).

10 Candidate Profile

- M.Sc. Computer Science, Universitas Indonesia (2016). Thesis: deep learning for biomarker discovery.
- Lecturer (Dosen Tetap), UBHINUS, since 2010. Courses: ML, NLP, Computer Vision.
- Published NLP work: Indonesian NLP survey (arXiv:2304.02746), low-resource dataset creation, sub-word segmentation, morphological analysis library (ModernKataKupas, PyPI).
- VOLCARCH project: 199 tracked experiments, 6 papers under review, arXiv preprint.
- Additional NLP portfolio: KorupsiNLP (693 court verdicts, sentencing prediction), NusantaraAgent (neuro-symbolic legal reasoning).

11 What This PhD Is and Is Not

This PhD *is*:

- An NLP thesis with contributions to historical text mining methodology
- Grounded in a concrete, physically validated archaeological application
- Designed to produce reusable tools and open datasets (Colonial Settlement Gazetteer, annotated historical Dutch NER corpus)

This PhD is *not*:

- An archaeology thesis (the archaeological interpretation is motivation, not contribution)
- A historical philology thesis (the candidate does not read 17th-century Dutch; the methodology is designed around this constraint)
- A repackaging of existing VOLCARCH work (all three papers are new; existing work is cited background)

References

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